

SCHEMA 1: FORMULATION

1. Design problem formulation

Explicit search bounds & constraints

- Swept bounds: PE array at most 1,024 PEs, L2 capacity 2, 3, or 4 MiB
- System obligations: 3 W TDP limit, 120 FPS frame rate constraint
- ISA & interconnect targets: RISC-V RV64GCV, UCle die-to-die, CXL expansion
- Technology evidence roles: predictive ASAP7 vs. foundry-qualified signoff

SCHEMA 2: LINEAGE

2. Attempt lineage header

Artifact identities & signed declarations

- Hashes identify prompt, parent commit, artifacts, configuration, and seed
- Multi-level IR representations: AST, CDFG, MLIR/CIRCT dialects
- Container image digest tracking: content digest of each tool image
- Tool CLI arguments & UPF (IEEE 1801) power domain state declarations

SCHEMA 3: CONTRACT

3. Environmental contract

Tool interface & behavior assessment

- Input prerequisites, output schemas, status semantics & error assumptions
- Assessed simulators & EDA tools: gem5, Verilator, SCALE-Sim, Ramulator, OpenROAD, Yosys, and OpenSTA
- Declared container capabilities: Protobuf state, checkpoint recovery, machine-parsable error channels

Selective metadata exchange under separate trust and access policy

Schemas make selected records interoperable; hashes and attestations identify artifacts and issuers rather than proving semantic correctness.
Access control, redaction, confidentiality review, and trust-root governance remain independent requirements.